PIC 10A 1A

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Today…

- C++Classes
  - Concepts, Examples, and Exercise Problems
- C++ Class Exercise: Wordle-ver. 1
  - (will be upgraded later)

C++ Classes

Concepts, Examples, and Exercise Problems

Exercise

- Consider the classes:

```
struct A {
    A();
    void a() const;
};
class B {
    void b() const;
public:
    B();
};
```

- with the accompanying code:

```
A x;
B y;
x.a();
y.b();
```

- Which statement generates a compiler error?

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Exercise (Answer)

- Consider the classes:

```
struct A {
    A();
    void a() const;
};
class B {
    void b() const;
public:
    B();
};
```

- with the accompanying code:

```
A x;
B y;
x.a();
y.b();
```

B::b() is a private method!
Thus cannot be accessed outside the class.

- Which statement generates a compiler error?

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Classes

- If a class has several data members, a proper initialization may be important

```
class B {
public:
    string name;
    double salary;
    int age;
    void b() const;
    B();
    //(another c'tor)
};
```

```
int main() {
    B b_object;
    B John("John Doe", 60000, 25);
}
```

Calling the default constructor here.
Setting the name = “”, salary = 0.0 and age = 0 in the default constructor can be a good initialization

Calling another type of constructor here
(not declared in the class interface yet)

What’s the correct signature of this constructor?

```
B(string _name, double _salary, int _age);
```

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Classes

- Always use the ***constructor initializer list*** for initialization

```
class B {
public:
    string name;
    double salary;
    int age;
    void b() const;
    B();
};
```

```
B(string _name, double _salary, int _age)
     : name(_name), salary(_salary), age(_age) {}
```

```
int main() {
    B b_object;
    B John("John Doe", 60000, 25);
}
```

The constructor's **body** is empty in this case

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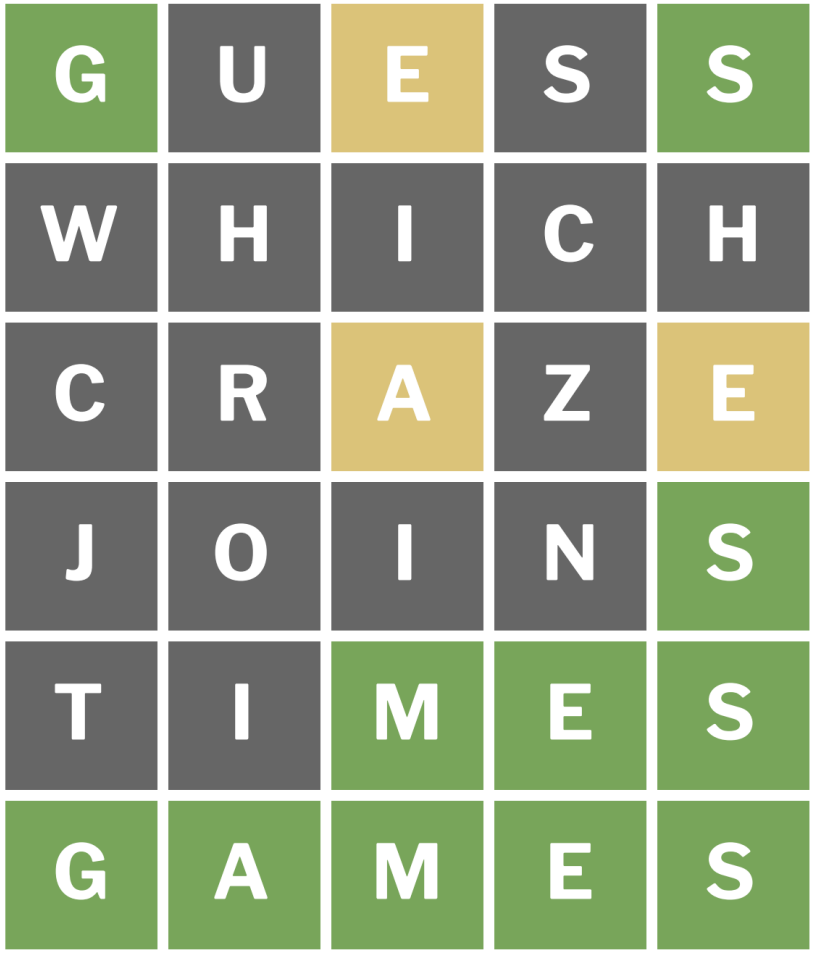
C++ Classes Exercise

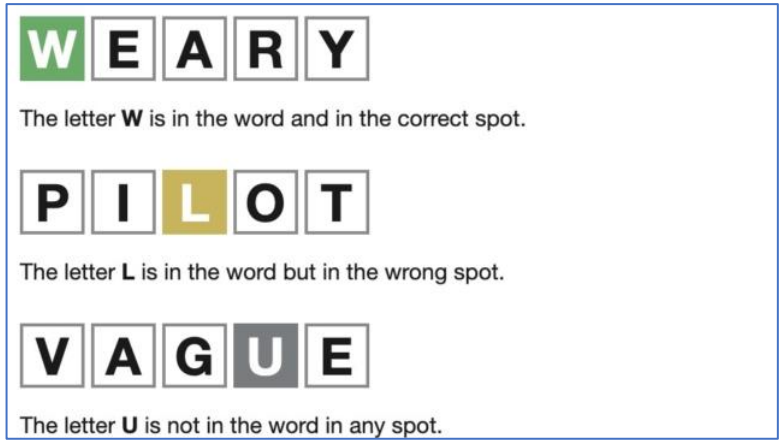
Wordle Ver. 1.0

Wordle

- Wordle is a word guessing games with three types of hints
  - You can actually play this game here: https://www.nytimes.com/games/wordle/index.html

- Three types of hints:
  - Green: correct letter + correct spot
  - Yellow: correct letter + wrong spot
  - Gray: wrong letter





Wordle

- We will write a class Wordle to play this game
- This class will have
  - a member “word”, which is the answer
  - a method “isCorrect()”, which returns true if the guess is correct
  - a method “giveHint()”, which gives a hint, where
    - green hints are “@”
    - yellow hints are “*”
    - gray hints are “-”
- Example game play:

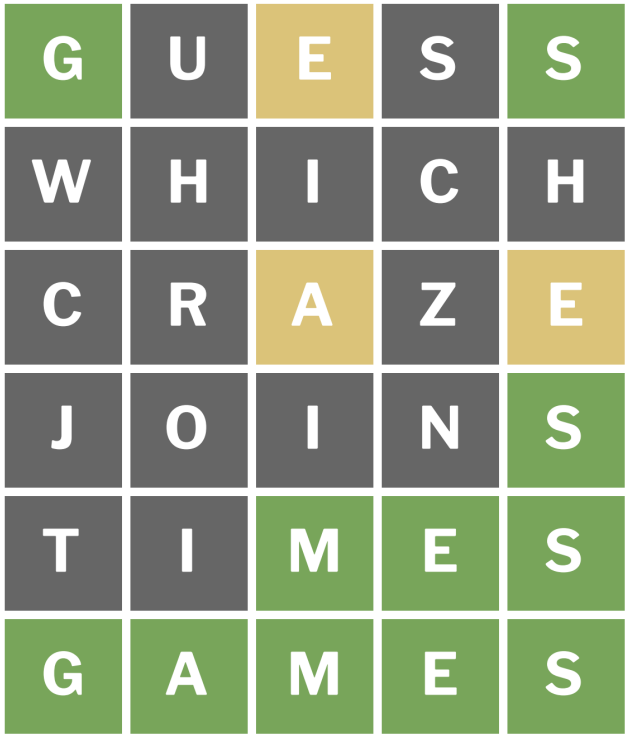
```
[1] Enter a 5-letter word :     abuse
                                @---@

[2] Enter a 5-letter word :     labor
                                **---

[3] Enter a 5-letter word :     price
                                *---@

[4] Enter a 5-letter word :     apple
                                @@@@@

Congratulations! You won the game in 4 rounds!
```



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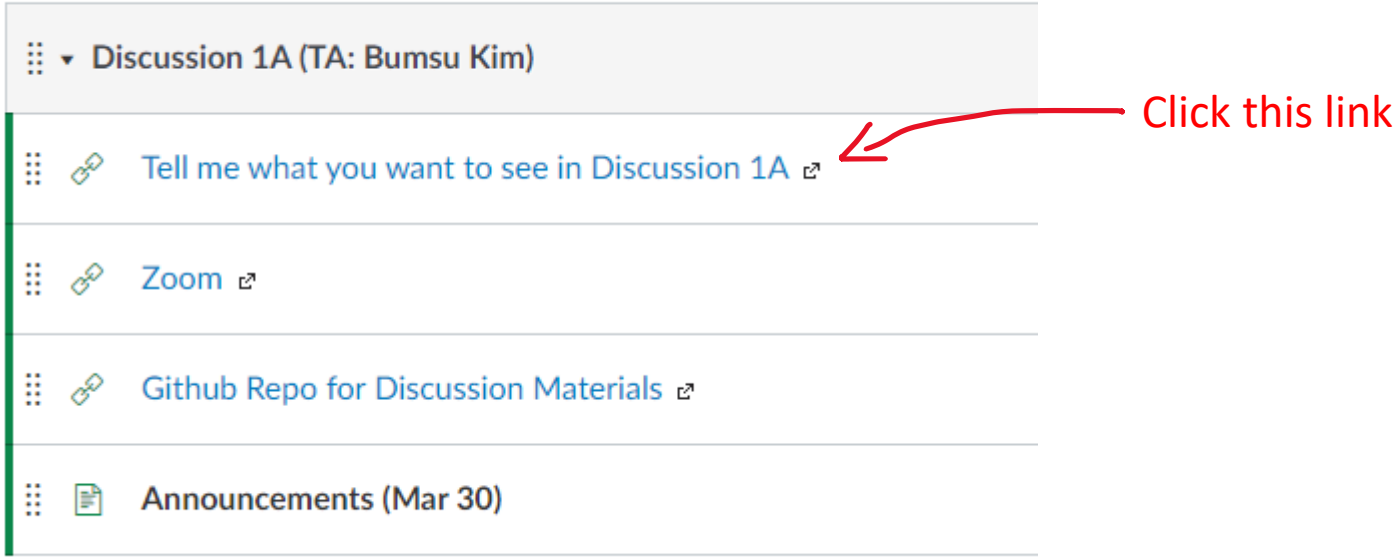
- Don’t hesitate to give a feedback on the discussion
- You can use a link in BruinLearn (Google Form)